Description

Illustration





Figure caption

1. The smoothness of concrete panels give the Sydney Opera house its iconic shape. © ANSYS, Inc. 2. The roof of the Pantheon in Rome, built nearly 2000 years ago, is still the largest unreinforced concrete dome in the world. Image used under license from Shutterstock.com

The material

Concrete is a composite, and a complex one. The matrix is cement; the reinforcement, a mixture of sand and gravel ('aggregate'), occupying 60-80% of the volume. The aggregate increases the stiffness and strength and reduces the cost (aggregate is cheap). Concrete is strong in compression but cracks easily in tension. This is countered by adding steel reinforcement in the form of wire, mesh or bars ('rebar'), often with surface contours to key it into the concrete; reinforced concrete can carry useful loads even when the concrete is cracked. Still higher performance is gained by using steel wire reinforcement that is pre-tensioned before the concrete sets. On relaxing the tension, the wires pull the concrete into compression; the concrete does not crack until the loads applied to it overcome this compression stress ('pre-stressed concrete'). Note that the properties quoted here refer to unreinforced concrete.

Compositional summary

0.6:1:2:4 Water:OPC:Fine:Coarse

General properties

Density		2.2e3	-	2.6e3	kg/m^3
Price	*	0.0302	-	0.0453	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓

Interiors	✖			
Services	✖			

Mechanical properties

Young's modulus	15	-	25	GPa
Shear modulus	* 6.5	-	10.9	GPa
Bulk modulus	* 7.1	-	11.9	GPa
Bending modulus	* 15	-	25	GPa
Poisson's ratio	0.1	-	0.2	
Yield strength (elastic limit)	1	-	1.2	MPa
Tensile strength	1.1	-	1.3	MPa
Compressive strength	* 13.3	-	30	MPa
Bending strength	1.7	-	2.4	MPa
Elongation	* 0	-	0.01	% strain
Hardness - Vickers	* 5.7	-	6.3	HV
Fatigue strength at 10^7 cycles	* 0.54	-	0.84	MPa
Fracture toughness	0.35	-	0.45	MPa.m^0.5
Mechanical loss coefficient (tan delta)	0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	0.385	-	0.606	m.°C/W
Thermal expansion coefficient	8	-	12	µstrain/°C
Specific heat capacity	835	-	1.05e3	J/kg.°C
Melting point	930	-	1.2e3	°C
Maximum service temperature	480	-	510	°C
Minimum service temperature	-160	-	-150	°C
Flammability	Non-flammable			
Emissivity	0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability	2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance	Average			

Electrical properties

Electrical conductor or insulator?	Poor insulator			
Electrical resistivity	1.85e12	-	1.85e13	µohm.cm
Dielectric constant (relative permittivity)	* 8	-	12	
Dissipation factor (dielectric loss tangent)	* 0.001	-	0.01	

Dielectric strength (dielectric breakdown)	0.8	-	1.8	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Castability	4	-	5	
Moldability	3	-	4	
Machinability	1			

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.117	-	0.129	kg/kg
Embodied energy (virgin grade production)	* 0.797	-	0.879	MJ/kg
Water Usage	* 2.65	-	2.93	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 16.4	-	19.8	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.07	-	1.3	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.0117	-	0.0129	kg/kg
Embodied energy, recycling	* 0.112	-	0.123	MJ/kg
Recycle fraction in current supply	14			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Concrete is a mixture of cement with fine and coarse natural aggregates and water, increasingly with specialist admixtures to impart improved properties; mortars and renders are based on cement, water and fine natural aggregate, and grouts are cement and water based.

Freshly-mixed concrete is fairly fluid, designed to achieve a specific consistency (e.g. slump). Poured into steel or timber molds ("sets" or "formwork") it can be shaped to form floors, walls and more elaborate structural elements. If these carry tension, steel or other forms of reinforcement must be used.

To achieve optimum performance, the concrete must normally be compacted, using vibrating pokers, to remove excess air from the mixing process and ensure the mix fully fills the corners of the formwork.

Concrete may not weather gracefully; unlike wood, stone and brick, unless it is detailed with care, it can stain, discolor and crack in a way that is visually unattractive. This can be overcome by cladding or painting the surface. Internally, there are few problems as concrete ages.

Concrete can either arrive at site in mixing trucks from a central batch manufacturing facility for pouring into moulds, or supplied as precast components (e.g. paving flags).

A special class is Self-compacting Concrete (SCC), which is formulated with smaller coarse aggregate sizes (typically 10mm maximum), and with cohesive and plasticising additives. Free-flowing concrete is produced that will self-level and needs no conventional vibro-compaction to consolidate it.
To achieve optimum performance, the concrete must normally be compacted, using vibrating pokers, to remove excess air from the mixing process and ensure the mix fully fills the corners of the formwork.

Technical notes

When the dry components of concrete are thoroughly mixed with water, the cement reacts to give hydrated compounds (C-S-H gel) that interlock and become solid, "gluing" the aggregates together into a strong, heterogeneous solid; a 100mm cube of concrete will typically support between 20 and 80 tonnes after 28-days, depending on the water/cement ratio used.

The exothermic hydration reaction between cement and water is initially slow to begin, so the mix remains fluid for around 90 minutes to 3 hours, allowing it to be transported to site and cast into formwork. Although the formwork can normally be removed after 24 hours, significant structural strength is not developed for 7 days (66-75% of ultimate), with full strength normally after 28-days, unless slower setting cements are used (see Cement).

Dense concrete made with Portland cement (CEM 1) is inherently durable once fully hydrated and endures wet environments and seawater, although it can be attacked by sulfates acids and selected chemicals. However, unreinforced concrete is rarely used, with most structural applications utilising Reinforced Concrete.

The quoted properties are for unreinforced concrete. However, in practice, they are significantly improved by reinforcement using steel bar or fibre.

Typical uses

General civil engineering construction and building.

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU & UK standards related to this material. Other countries have their own standards: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity
BS 8500: Concrete. Complementary British Standard to BS EN 206
CSI/CSC 03 33 13: Heavyweight Architectural Concrete

Links

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Reference

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